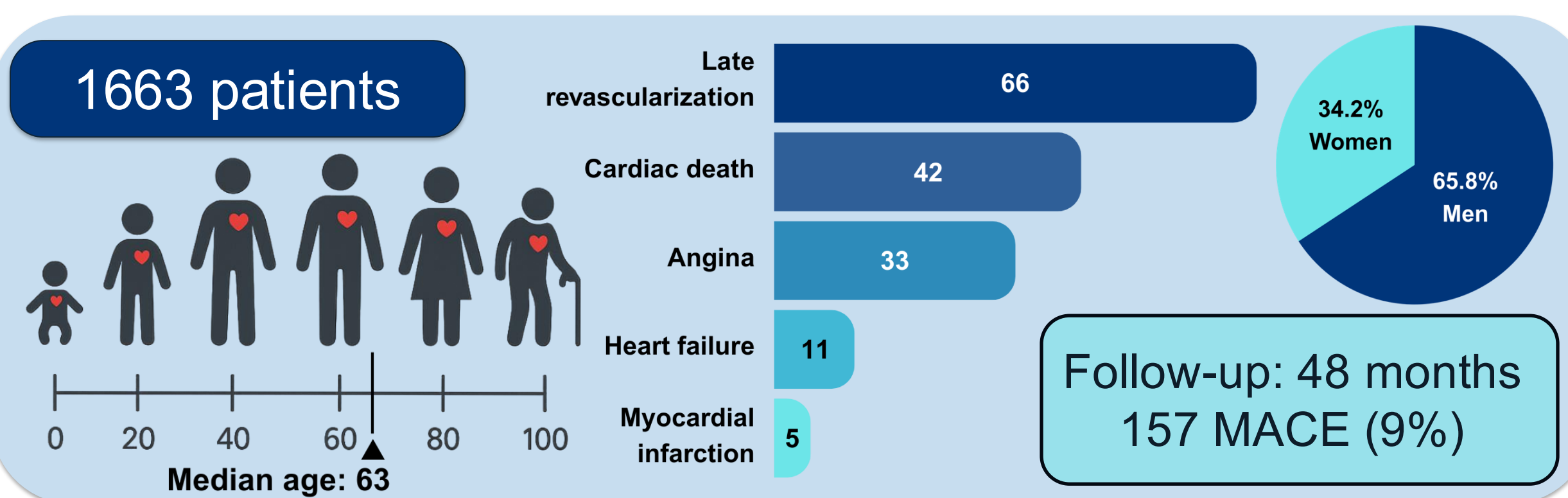
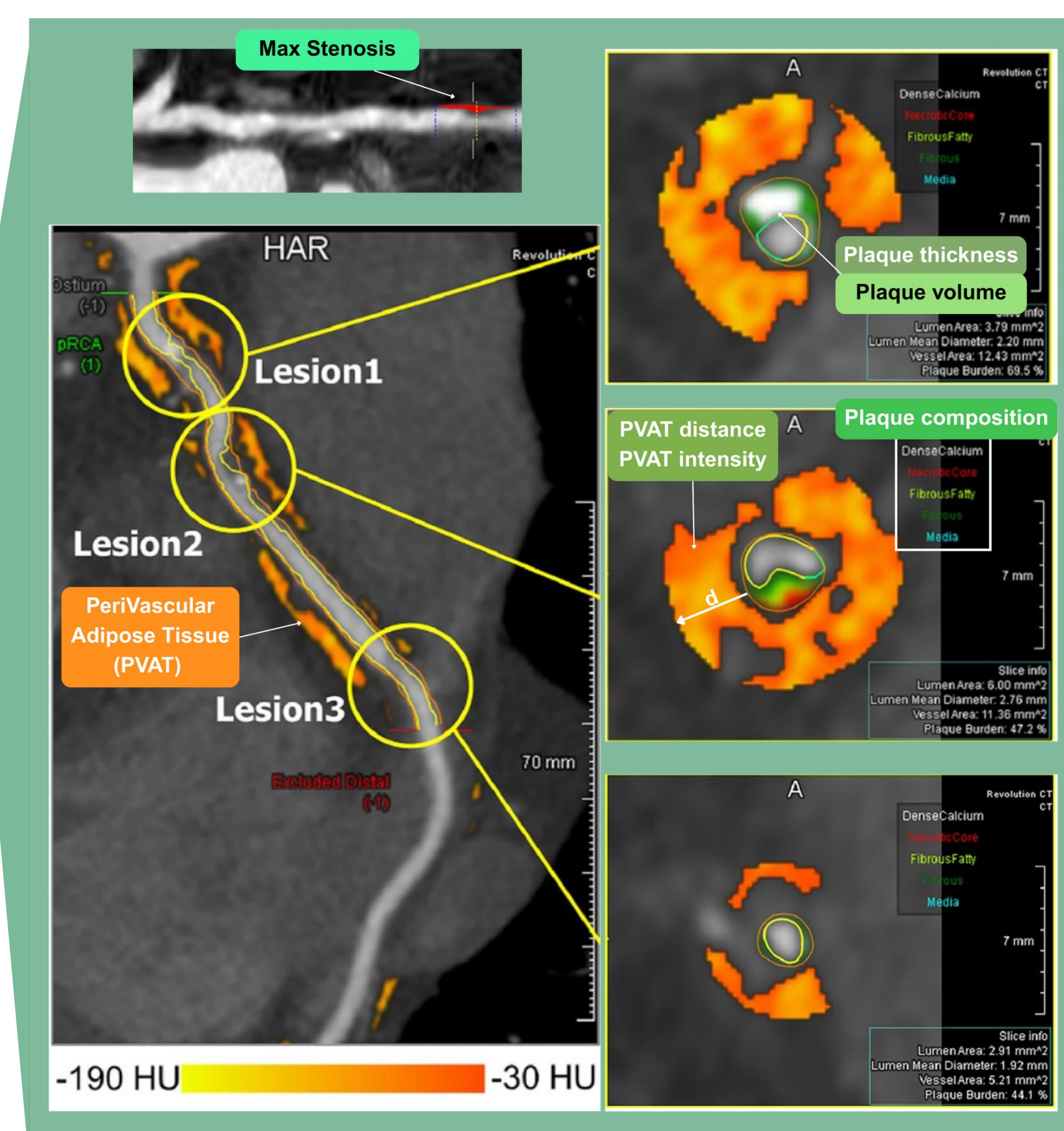
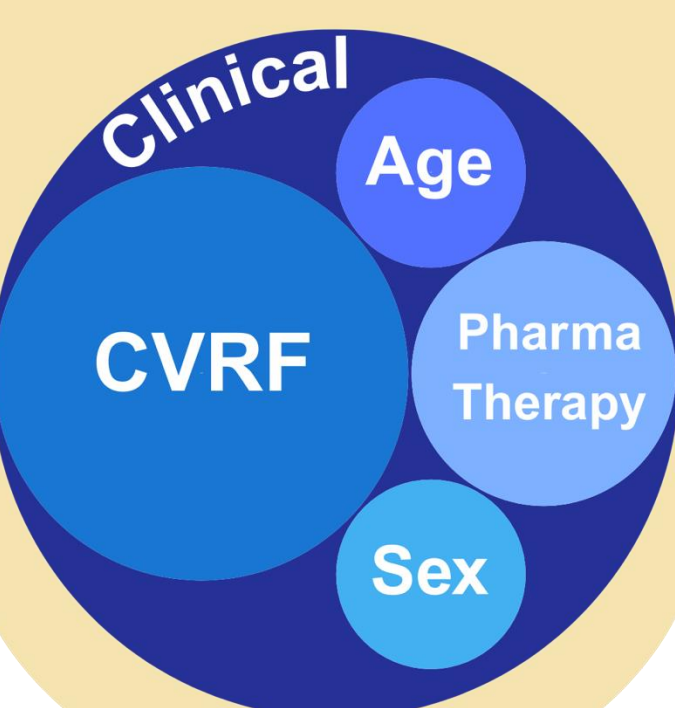
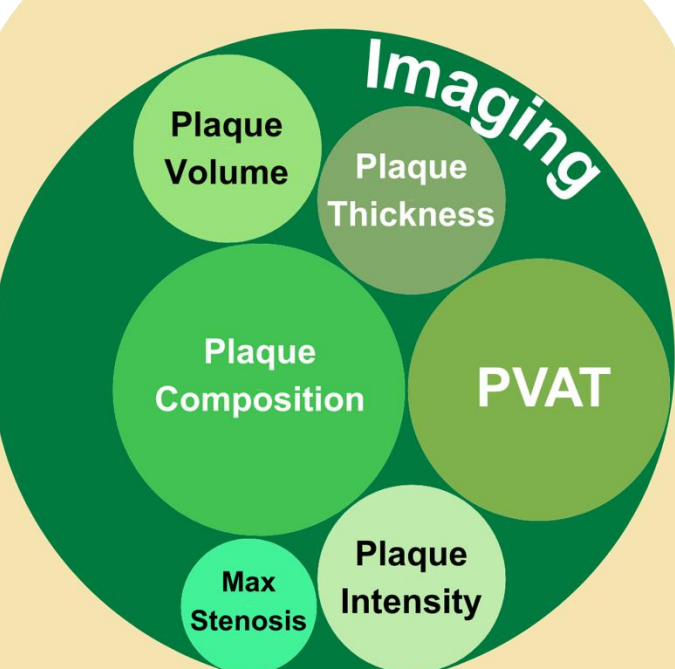


Classification of CAD: state of the art

- CAD is often classified with **semi-quantitative scores** (e.g., CAD-RADS 2 [1]) relying on **standard features** such as stenosis severity, plaque composition and adverse plaque characteristics;
- These **overlook rich quantitative data** from CCTA;
- Dedicated software can extract these **quantitative features**;
- Leveraging them may **improve risk prediction** and enable **more precise patient stratification**.

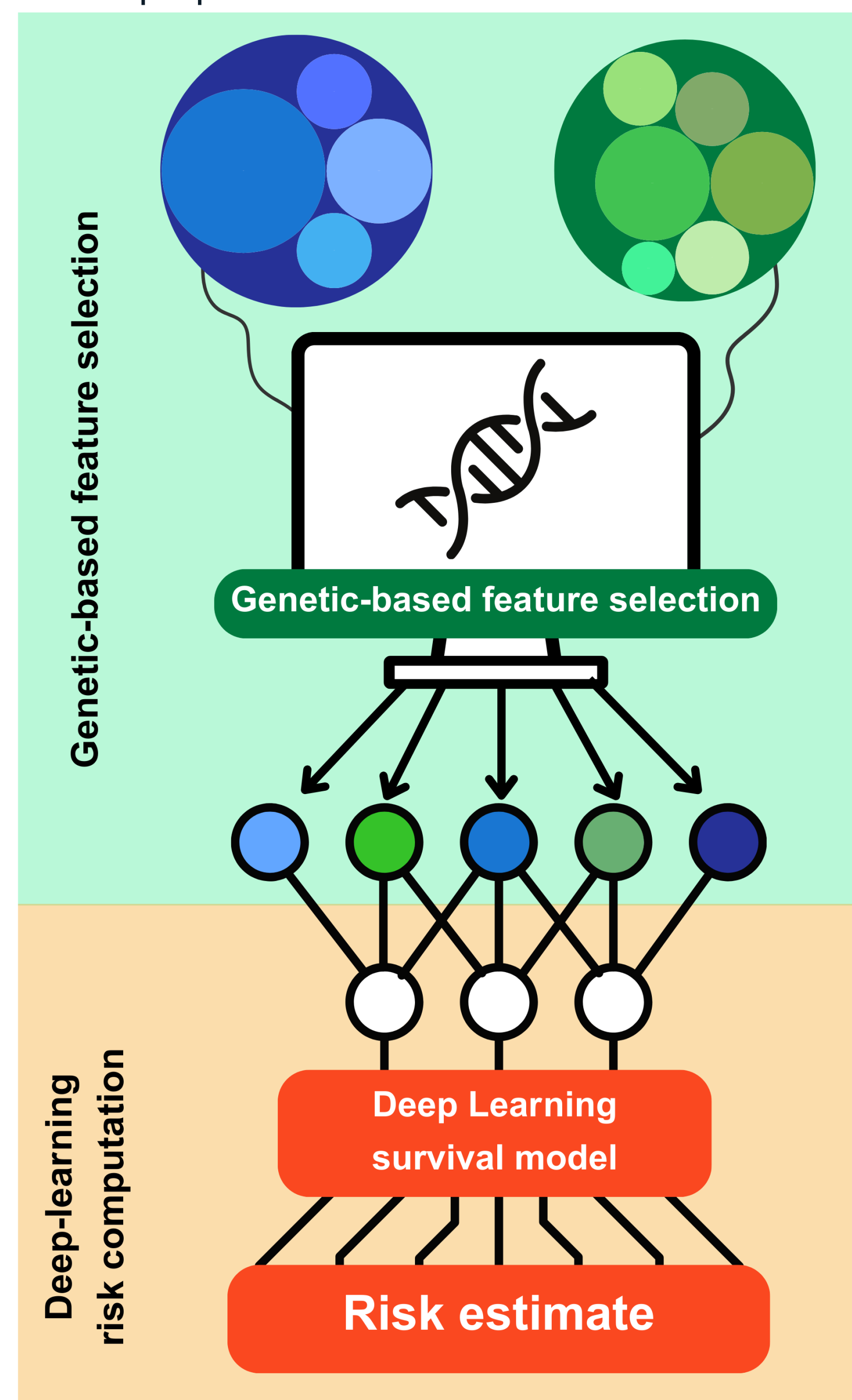


Variables

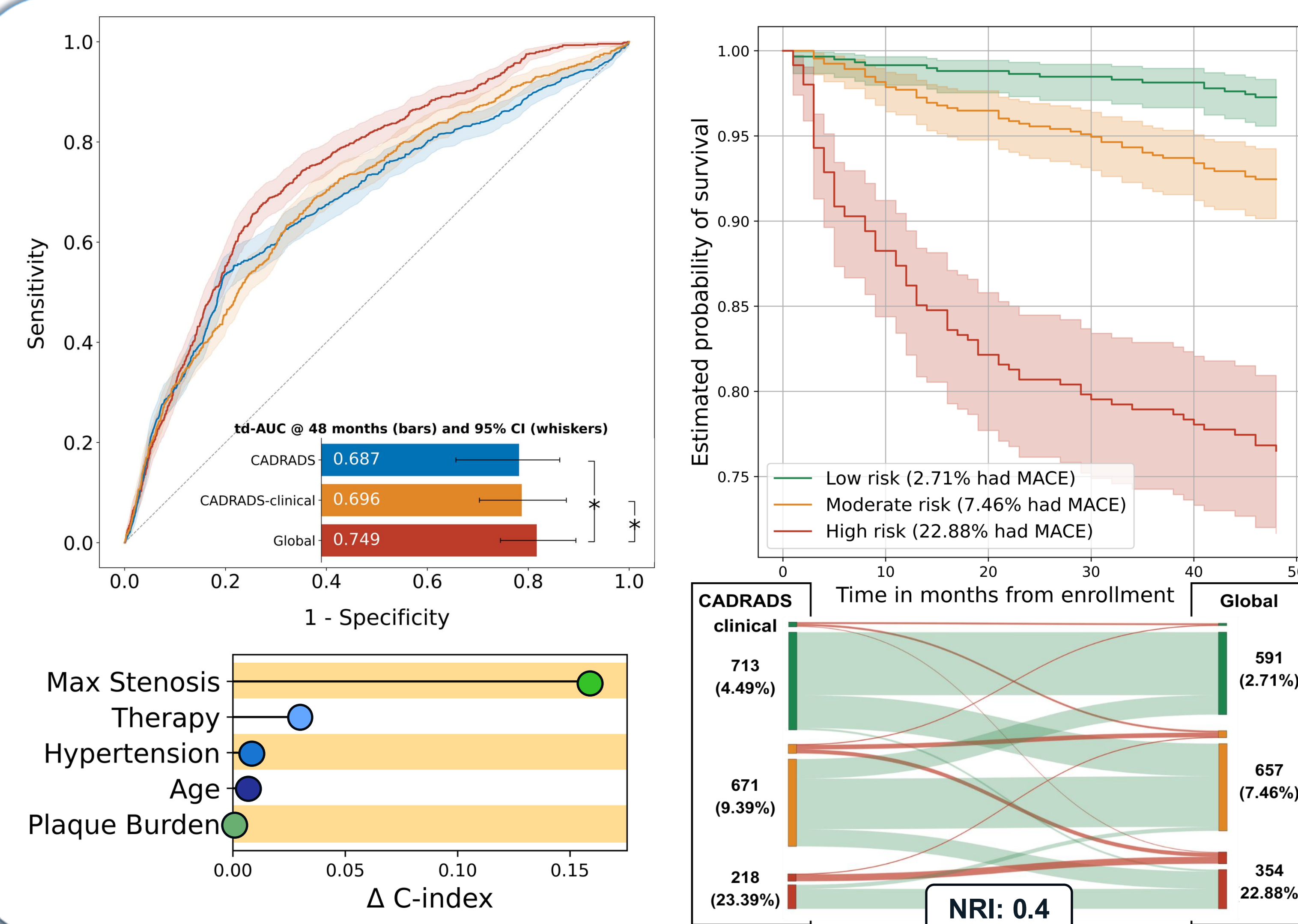


AI-based risk computation

- Data-driven** feature selection for most informative variables with **genetic-based algorithm** → optimize risk prediction through an evolutionary strategy;
- Deep learning** [2] survival model → get over traditional Cox model's assumptions on proportional hazards.



Model performance and patients' stratification



Conclusions

Quantitative features from CCTA improve risk prediction and patient stratification. Integrating these imaging biomarkers with clinical and pharmacological data enables a more complete and accurate assessment. Although **data-driven**, the selected features remain **clinically interpretable**, with **maximal stenosis** being consistently the strongest predictor.

References

- [1] CAD-RADS™ 2.0 - 2022 Coronary Artery Disease-Reporting and Data System. Cury, Ricardo C. et al. Journal of Cardiovascular Computed Tomography, Volume 16, Issue 6, 536 – 557; [2] Deep Survival Machines: Fully Parametric Survival Regression and Representation Learning for Censored Data With Competing Risks. C. Nagpal et al. IEEE Journal of Biomedical and Health Informatics, vol. 25, no. 8, pp. 3163-3175, Aug. 2021.